

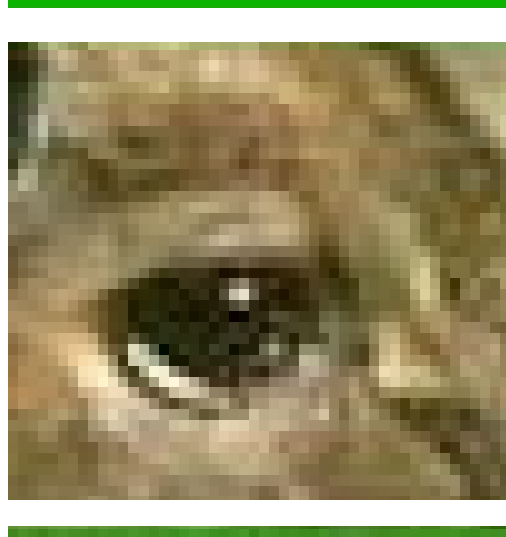
Rasters

A raster is a grid data structure, which represents an image as a matrix of equal sized squares or cells. Each cell is called a picture element or *pixel*. Each pixel represents a certain amount of ground area. These pixels can be extremely small so as to give the impression of a normal picture. Upon magnification one can see the matrix pattern. Each of these pixels can have a code or classification that tells us something about that pixel. For example, each pixel in an image of the earth's surface will have a value code that tells us what kind of land cover it represents.

Each raster has a specific *resolution*, which means the pixels are all of a single size, which can range from very small (which appears as smooth lines) to very large (which gives us a rough image). Note the eye of the goat on the left below appears as a group of small boxes forming the shape of the eye in the image on the right.



at full resolution



at 1/10th resolution

The raster image of Canada that will be used is an AVHRR (Advanced Very High Resolution Radiometer) satellite image. As the name depicts, advanced very high resolution implies that a very high number of pixels were used to give a clear picture of Canada's land cover. To learn more about AVHRR imagery, access the Canada Centre for Remote Sensing web site at: <http://www.ccrs.nrcan.gc.ca/ccrs/eduref/tutorial/indexe.html>

An image at a 1: 2 million scale means that one unit on the map surface represents 2 million units on the earth's surface. Therefore, if we measure 1 centimeter on the map, it will represent 2 million centimeters, or 20 kilometers on the earth's surface. The AVHRR pixels represent 1 km. by 1 km. square, therefore, at that scale a single pixel measures just 0.05 cm. by 0.05 cm.

The AVHRR image that we will use has pixels that are several times larger in order to speed up processing time. The image is classified into 14 land cover classes and each class has a class value that is between 0 (black) to 255 (white). The table below gives these class values and a brief description of the land cover of each class.

AVHRR Legend

Value	Legend text	Description
0	Urban	Built up areas
12	Coniferous forest	Forest consisting primarily of coniferous trees
30	Taiga	Stunted coniferous forest lying between tundra and boreal
31	Snow & ice	Permanently covered in snow or ice, glaciers
90	Mixed coniferous	Mainly coniferous forest with some deciduous trees
146	Tundra	Treeless, mainly marshy region with underlying permafrost
168	Deciduous forest	Forest consisting primarily of deciduous trees
175	Shrubland & alpine	Low growing shrubs, heather, moss and lichen
176	Mixed deciduous	Mainly deciduous forest with some coniferous trees
219	Barrens	Bare soil and rock supporting only moss or lichen
224	Forest burns	Areas where forest fires have occurred
248	Grassland	Land with herbaceous (non-woody) vegetation
252	Agriculture	Land primarily used for raising crops or livestock
255	Water	Large areas of open water

When you use the AVHRR image in MFWorks, you will be asked to classify the values in a *legend*, which identifies each colour on the map as a land cover class. You will do this by matching the value in the legend to the table above and entering the legend text in its place. Later you will be asked to combine some of the 14 classes to give you 10 new classes. This will be done by combining the 5 forest classes of deciduous (2), coniferous (2) and burns into one class.

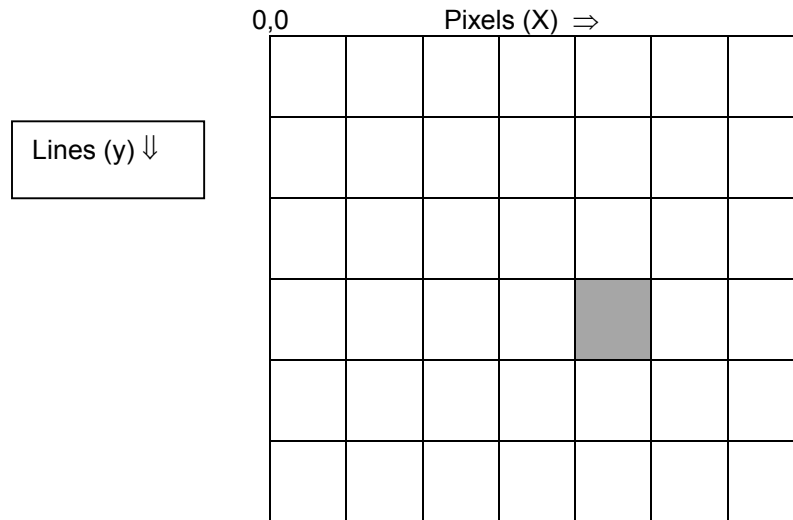
Rasters Student Activity Sheets

Name: _____

In Remote Sensing, the coordinate origin is at the upper left of the image. The X value is the number of pixels across the image, and the Y value is the number of lines down. The following diagram illustrates how a pixel coordinate value is determined.

X = 7 pixels across

Y = 6 lines down



The coordinate of the shaded pixel is (5, 4).

On the next page, you will find a representation of an 8-bit image of 256 levels from 0 to 255, where 0 is black, and 255 is white. Usually, remotely sensed data is collected as 8 or 16-bit data: 8-bits for optical images, 16-bits for radar images like the images of our Canadian Earth Observation satellite RADARSAT. Computers read bits, or binary digits, consisting of a value of 1 or 0. A single bit image consists of pixels that can only be displayed as black or white (1 bit = 2^1). An 8-bit image (2^8) has 256 Grey levels that could be represented by different colours. A 16-bit image (2^{16}) has 65536 Grey levels that could also be represented by different colours but becomes more complex when in colour. In order to represent each pixel in colour on a computer, we need to add values to the different Red, Green and Blue Guns. Here is an example of how a computer creates colours.

Red Gun	Green Gun	Blue Gun	Colours we see on screen
0	0	0	Black
255	0	255	Magenta
0	255	255	Cyan
255	255	0	Yellow
255	255	255	White

Look closely at the computer monitor or at a TV screen with a magnifying glass. You will see clusters of red, green and blue dots. Each cluster represents a pixel and depending on the intensity or brightness of each dot in the cluster a specific colour is seen for that pixel. In the cathode ray display, the red dots are controlled by the red "electron gun" the green dots are controlled by the green "electron gun" and the blue dots are controlled by the blue "electron gun". If you look at different coloured areas on the screen with the magnifying glass you will see different intensities of light.

On this 8-bit image where each pixel is 10m x 10m, create the Grey scale value of each pixel to find the final image. Give also the coordinate of all the pixels that is not white.

255	155	255	155	255	155	255
255	255	255	255	255	255	255
255	255	0	255	0	255	255
255	50	255	255	255	50	255
255	255	255	0	255	255	255
255	0	255	255	255	0	255
255	255	0	0	0	255	255

Pixels coordinates

_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

FILL IN THE ECOZONE NAMES

